Drift-Aware Reference Shrinkage for Proactive Traffic Engineering with Generative Network Twins

Jasmine Kaur
Yogananda School of AI, Computers and Data Sciences
Shoolini University of Biotechnology and Management Sciences
Solan, Himachal Pradesh, India
jaledarjasmine@gmail.com

Bashar Aswad Kokaz Al-Saeedi
Yogananda School of AI, Computers and Data Sciences
Shoolini University of Biotechnology and Management Sciences
Solan, Himachal Pradesh, India
basharenas990@gmail.com

Alok Raj*
Yogananda School of AI, Computers and Data Sciences
Shoolini University of Biotechnology and Management Sciences
Solan, Himachal Pradesh, India
alokraj2027@gmail.com
ORCID: 0009-0003-7870-7065

Pankaj Vaidya
Professor
Yogananda School of AI, Computers and Data Sciences
Shoolini University of Biotechnology and Management Sciences
Solan, Himachal Pradesh, India
pankaj.vaidya@shooliniuniversity.com
ORCID: 0000-0003-1304-630X

Anurag Rana
Professor
Yogananda School of AI, Computers and Data Sciences
Shoolini University of Biotechnology and Management Sciences
Solan, Himachal Pradesh, India
anuragrana.anu@gmail.com
ORCID: 0000-0003-0247-8908

**Corresponding author*

***Abstract*—Proactive traffic engineering forecasts the traffic matrix one control interval ahead and optimises routing against the forecast. It replaces a measurement with a prediction, and is therefore profitable only under a condition rarely tested: that the one-step forecast error is smaller than the traffic's drift over the same interval. The condition is measured from telemetry alone and test it by varying the latent rank, which moves the forecast error while leaving the traffic, and so the drift, unchanged. A Generative Network Twin is developed, a conditional LSTM trained with the energy score, and couple it to a predict-then-optimise controller through a twin–reality drift gate: an online statistic that tracks how far the twin has departed from measurement and shrinks the optimiser's reference towards live telemetry. On a queueing-accurate NSFNET simulator across one in-distribution week and five unseen scenarios, the full MSE+adversarial+energy-score objective raises coverage of nominal 90% intervals from 2.8% to 74.7% and lowers CRPS by 25.9%, though undercoverage remains. The condition does not predict the in-distribution sign, but it predicts the penalty under shift: when it holds the raw forecast costs 0.9% mean delay and loses on one of six scenarios, and when it fails, 8.9% and four of six. The gate recovers the loss in both regimes. Two null results are also reported: the adversarial term contributes little consistently alongside a proper scoring rule, and our PPO agents converge to static OSPF and lose to classical local search.**

***Index Terms*—network digital twin, traffic matrix forecasting, energy score, calibrated uncertainty, proactive traffic engineering, distribution shift**

I. Introduction

A backbone operator faces a control problem with an awkward timing property. Congestion builds over seconds to minutes, and measuring it, deciding on a response, and pushing new interior gateway protocol (IGP) weights to routers takes a comparable amount of time. By the time a reactive controller has responded, the event has often passed, or the response is now mistuned for the traffic that has since arrived.

The proposed remedy is proactive control: forecast the traffic matrix one interval ahead, optimise routing against the forecast, and install the configuration before the traffic arrives. This is attractive, widely proposed [1]-[3], but conditionally fragile.

The fragility has a simple source. Proactive control replaces a measurement with a prediction, which pays only when the prediction is more accurate than the measurement it displaces. Writing ε_{fcst} for the relative one-step forecast error and δ_{drift} for the relative change of the traffic matrix over one control interval, proactive control on a raw forecast is worth doing only when

$$\varepsilon_{fcst} < \delta_{drift}. \tag{1}$$

At the five-minute granularity typical of backbone telemetry this is far from automatic, and it is seldom evaluated explicitly. Under distribution shift, when a network enters a regime absent from the training data, the forecaster's error grows precisely when accurate control matters most, and a forecaster trained by mean-squared error gives no signal that this has happened.

A. Contributions

1) A measurable criterion, and what it does and does not predict. Equation (1) is stated in a form computable from telemetry alone and evaluated by varying the latent rank k, which raises the forecast's error floor while leaving the traffic — and therefore δ_{drift} — untouched. The criterion does not cleanly predict the in-distribution sign: the raw forecast beats reactive routing in distribution on both sides of the inequality. What it predicts is the penalty once the network leaves the training regime. With the inequality satisfied the raw forecast costs 0.9 % mean delay across six scenarios and loses on one of

them; with it violated, 8.9 % and four of six, and its worst case degrades from $+18\,\%$ to $+77\,\%$. This distinction is retained because the weaker claim is the one the data supports.

2) Calibration through a proper scoring rule. A conditional LSTM twin is trained with a composite objective including the energy score [4], a strictly proper scoring rule for multivariate predictive distributions, applied here to traffic-matrix forecasting. It raises empirical coverage of nominal 90 % intervals from 2.8 % to 74.7 % and reduces CRPS by 25.9 % at a cost of 0.9 % in MAE. In particular, 74.7 % coverage is substantially improved but still under-calibrated.

3) The twin–reality drift gate. An online statistic is defined ρ_t that tracks the discrepancy between what the twin predicted and what was subsequently measured, and use it to interpolate the optimiser's reference matrix between forecast and live telemetry. This shrinkage estimator is what makes the twin safe to rely on: under severe shift it degrades exactly to the reactive controller rather than to something worse.

Two negative findings are reported: adversarial training contributes nothing measurable once a proper scoring rule is present, and the evaluated deep reinforcement learning (DRL) agents converged to the static OSPF operating point and were beaten by classical local search.

II. Related Work and Positioning

Traffic matrix estimation and forecasting. The low effective rank of backbone traffic matrices is a long-standing empirical finding [5], and motivates the dimensionality reduction used here. Deep-learning approaches have replaced linear predictors with recurrent [1], [2] and graph-convolutional [3] architectures, generally reporting improvements in point accuracy measured by MAE or RMSE. That near-universal use of point-accuracy metrics is precisely the gap examined here: a forecaster can dominate on MAE while being useless for risk-aware control, because MAE says nothing about whether the model knows when it is wrong. The results quantify this directly.

Generative models for network traffic. Generative adversarial networks [6] aim to learn structured data distributions, including temporal variability. The ablation provides a contrasting result. Proper scoring rules [7] provide the alternative used here: the energy score [4] generalises CRPS to multivariate targets and is strictly proper, giving a principled objective for ensemble forecasting that adversarial training approximates only indirectly.

Network digital twins. Twin architectures for networking [8], [9] propose a continuously updated model used for what-if analysis and control, and most treat the twin's fidelity as a design assumption. The proposed drift gate takes a different approach: it treats twin fidelity as an online-measured quantity and modulates the controller's reliance accordingly. Section VII shows this is not a refinement but the difference between the proactive controller helping and hurting under shift.

Learning-based traffic engineering. Reinforcement learning for routing has been explored with action spaces from per-flow path selection to link-weight manipulation [10], [11]. Reported gains vary widely and independent reproduction is uncommon. Classical local-search traffic engineering [12] remains a strong and under-reported baseline; the present results support that assessment. Robustness to distribution shift is a general concern in learned systems [13].

III. System Model and the Criterion

A. Network and performance model

The network is a directed graph $G=(V,E)$ with $N=|V|$ nodes and $L=|E|$ directed links; link e has capacity c_e and propagation delay v_e. Demand is a traffic matrix vectorised as $d_t \in R_{+¿^{P}¿}$ over $P=N(N-1)$ origin–destination pairs. Forwarding is destination-based shortest path under positive weights w, which is what an IGP implements. With $A(w)\in\{0,1\}^{L\times P}$ the path-incidence matrix, link load is linear in demand:

$$y=A(w)d,\ u_e=y_e/c_e. \tag{2}$$

Equation (2) is the key computational device. Building $A(w)$ costs N Dijkstra runs; afterwards, evaluating any number of demand matrices under that routing is one matrix product. Without this, the ensemble and CVaR methods below would be out of reach rather than merely slow.

Per-link queueing delay follows the M/M/1 waiting time [14] with mean burst service time s, saturated at q_{max}, and loss composes M/M/1/K blocking with hard overflow:

$$q_e=min(s\,u_e/1-u_e,q_{max}),\ \delta_e=v_e+q_e. \tag{3}$$

End-to-end delay sums along paths and survival multiplies, giving demand-weighted mean delay $\acute{D}$ and loss $\acute{L}$. The control cost blends the two with an overload penalty:

$$c(w,d)=\frac{\acute{D}(w,d)}{D_0}+\lambda_L\acute{L}(w,d)+\frac{\lambda_O}{L}\sum_e max(0,u_e-1). \tag{4}$$

The controller picks w_t using only information available at the start of interval t, and is scored on $c(w_t,d_t)$ under the realised d_t. This one-interval information gap is the entire source of difficulty.

B. Latent representation and the error floor

Traffic is multiplicative and approximately low rank, so the model works in log space on a PCA basis of rank k: $x_t=logmax(d_t,d_{min})$, $z_t=S^{-1}U^{\top}(x_t-\acute{x})$, decoding as $\hat{d}_t=\exp(US\,z_t+\acute{x})$. Encoding then decoding is lossy, and that residual is an irreducible floor on every downstream forecast: no predictor operating in this latent can beat it on the traffic matrix. It is reported in the same relative L_1 norm used throughout for reference error and drift, so that the three quantities are comparable. At $k=16$ the floor is $11.4\,\%$ and

rank 16 explains 92.1% of log-space variance; at $k=8$ the floor rises to 21.0%. The traffic's one-interval drift is 16.8% and depends on k not at all, which is what makes k a clean instrument for moving one side of (1).

This floor is what makes (1) a design constraint rather than a curiosity. Because ε_{fcst} is bounded below by representation loss, the useful engineering lever is often not a better sequence model but a richer representation — and, as Section VII shows, the rank that fixes the floor also fixes how badly proactive control degrades under shift.

IV. The Generative Network Twin

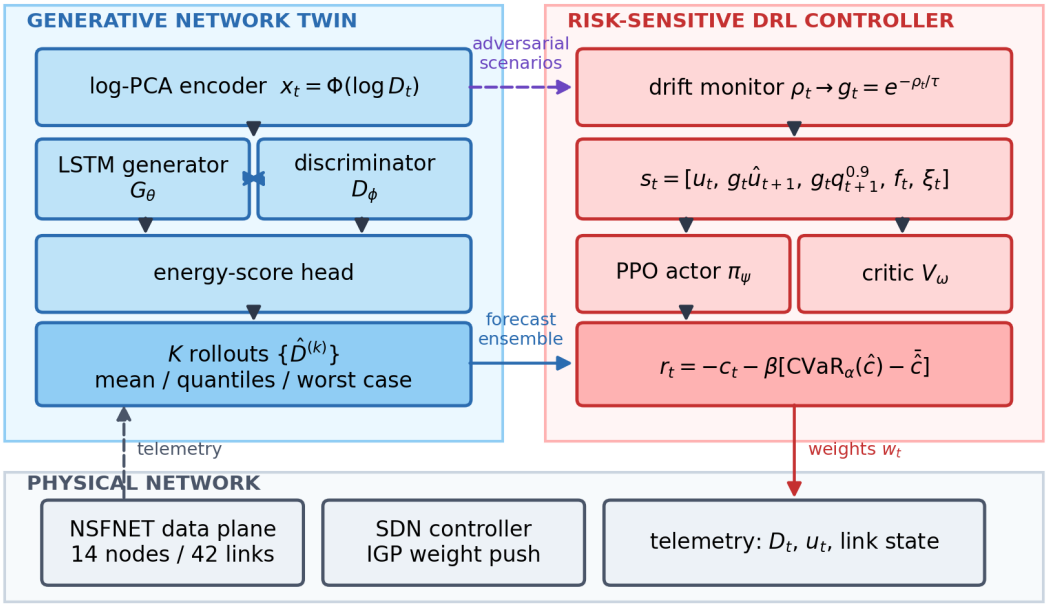


Fig. 1. System architecture. Telemetry feeds the generative twin, which emits an ensemble of candidate future traffic matrices; the drift monitor measures how far the twin has departed from reality and gates the controller's reliance on it.

The generator is a single LSTM [15] of hidden width 64 conditioned on a window of L_{win} observed latents, decoding H_f steps in free-running mode. At every step the input is the concatenation $[\zeta_t; \epsilon_t]$ with fresh noise $\epsilon_t \sim N(0, I)$, making the model a stochastic forecaster rather than a deterministic map. Two design choices affected convergence. The output head is residual, $\hat{z}_{t+1} = \zeta_t + W_o h_t + b_o$, so with W_o initialised to zero the model begins training as exact persistence; without this it failed to reach persistence-level accuracy within budget. Training uses scheduled sampling [16], annealing the teacher-forcing probability from 0.70 to 0.05.

The generator minimises

$$L_G = \lambda_m L_{mse} + \lambda_a L_{adv} + \lambda_e ES. \tag{5}$$

Used alone, the MSE term produces the conditional mean, which by construction has no predictive spread; this is the origin of the calibration failure observed, and it is a property of the objective, not the architecture. The adversarial term is the non-saturating generator loss against a three-layer MLP discriminator. The third term is the energy score, estimated from two independent rollouts $\hat{Z}, \hat{Z}'$ sharing conditioning but differing in noise:

$$ES = 1/2\left(\|\hat{Z} - Z\| + \|\hat{Z}' - Z\|\right) - 1/2\,\|\hat{Z} - \hat{Z}'\|. \tag{6}$$

Strict propriety means (6) is uniquely minimised in expectation when the predictive distribution equals the true conditional. The first term rewards accuracy; the second rewards spread, directly counteracting the variance collapse associated with MSE. Its gradients are differences of unit vectors, costing one extra forward and backward pass per iteration.

V. The Twin–Reality Drift Gate

The gate evaluates whether recent twin forecasts remain consistent with observed telemetry. At interval t the controller compares what the twin predicted against what was measured, normalise by volume, and smooth exponentially:

$$e_t = \frac{\|d_t - \hat{d}_{t\,|\,t-1}\|_1}{\|d_t\|_1}, \rho_t = (1-\beta)\rho_{t-1} + \beta e_t. \tag{7}$$

This costs one vector subtraction per interval and needs no ground truth beyond telemetry the controller already collects. This produces a trust gate

$$g_t = \exp(-\rho_t/\tau) \text{ if } \rho_t \le \rho_{max}, \text{else } 0. \tag{8}$$

Given ensemble $\{\hat{d}^{(j)}\}_{j=1}^{K}$, let T index the $\lceil \alpha_r K \rceil$ members with highest cost under the current routing, and form

the risk-weighted reference $R_t = \frac{1}{2}\bar{\hat{d}} + \frac{1}{2}|T|^{-1}\sum_{j \in T}\hat{d}^{(j)}$, a conditional value-at-risk construction [17] that biases the optimiser towards configurations surviving the ensemble's unfavourable tail. This term is meaningful only if the ensemble is calibrated, which is exactly what the energy score buys: (6) and (9) are two halves of one mechanism. The optimiser's reference is then

$$d_t^{ref} = g_t R_t + (1 - g_t) d_{t-1}. \quad (9)$$

Equation (9) is a shrinkage estimator trading the forecast's bias against telemetry's staleness, with the mixing weight estimated online. When the twin is trusted $g_t \to 1$ and control is proactive; when discrepancy exceeds the threshold $g_t \to 0$ and the reference moves toward available telemetry.

The optimiser itself is a deliberately classical Fortz–Thorup-style randomised descent [12] over $N_{¿}$ proposals, warm-started from the previous interval's weights. Holding it, its budget and its seed fixed across every controller variant means any performance difference is attributable solely to d^{ref}.

For comparison, the study also trains a risk-sensitive DRL controller (GNT-PR) using PPO [18] with GAE-λ [19] over a diagonal Gaussian policy that multiplicatively perturbs OSPF weights, $w = w_{ospf} \odot \exp(\kappa \tanh a)$, with a baseline-relative reward, a CVaR risk premium and a churn penalty. All networks are optimised with AdamW [20].

VI. Experimental Setup

Traffic is synthetic, and the generation process is described to expose its assumptions. Node populations are log-normal and the base matrix is gravity; each interval applies diurnal and weekly seasonality with per-node time-zone offsets, node-level Ornstein–Uhlenbeck drift entering multiplicatively on both endpoints, idiosyncratic log-normal noise, and Pareto-amplitude flash crowds with a gradual onset. Aggregate demand is calibrated so static OSPF sees a mean maximum link utilisation of 0.78, a regime where traffic engineering has room to act but the network is not permanently saturated. Training uses 40 independent weeks; validation, test and DRL weeks are held out separately, so the agent never trains on the evaluation trace. The reported held-out evaluation spans 400 control intervals per scenario; the training duration is 40 independent weeks. Exact validation-set and DRL trace sizes are not provided in the available experiment summary; therefore, comparison across these partitions remains limited.

Five out-of-distribution scenarios, none represented in training: S1 flash crowd (three pairs surge $\times 8$); S2 distribution shift (new gravity draw, doubled burst rate, so the PCA basis is mis-specified); S3 volumetric surge (traffic to two sink nodes amplified); S4 overload ($\times 1.45$ global demand); S5 dual link failure.

Everything is dependency-free NumPy with hand-derived gradients. Numerical verification finds that LSTM backpropagation through time agrees with central differences to a maximum relative error of 2.2 x 10^-7 (including the scheduled-sampling path, where a teacher-forced step must cut the gradient into its input); the energy-score gradient to 2.7 x 10^-8; the vectorised path-matrix evaluator matches a naive per-path reference to approximately 10^-15; and PPO converges on the known optimum of a synthetic control task, improving from -0.64 to -0.12 against an optimum of 0. Forecast metrics use K = 100 evaluation rollouts; the controllers use K = 8, so the CVaR tail contains two members. Energy-score variants train for 3,000 iterations against 4,000 for the others, each iteration performing two generator passes, so the ablation aims to match computational cost rather than iteration count. The implementation companion, configuration files, and scripts are available at https://github.com/jasminebug/-drift-gated-generative-network-twins.

VII. Results

A. Calibration: the failure and the fix

TABLE I

Twin variants on the held-out test week (K = 100). Lower is better except coverage; RMSE tracks MAE and is omitted.

Variant	MAE	CRPS	Cov@90	W_1
Persistence	0.3789	0.3789	—	—
LSTM (mse)	0.3437	0.3377	2.8 %	0.1216
GAN (adv)	0.5341	0.4798	21.5 %	0.0424
T1 (mse+adv)	0.3695	0.3226	21.7 %	0.0605
T2 (mse+es)	0.3460	0.2526	70.7 %	0.0238
GAN-LSTM	0.3468	0.2501	74.7 %	0.0100

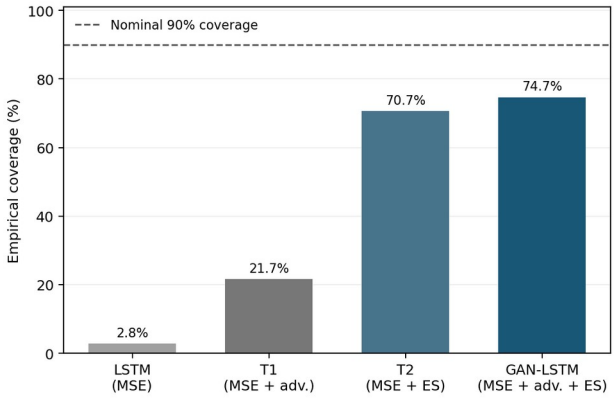


Fig. 2. Empirical coverage at the nominal 90% interval level from Table I. The energy-score variant improves coverage, but all variants remain below 90%.

The main calibration result appears in Table I. An MSE-trained LSTM — the standard choice in the traffic-forecasting literature — produces 90 % predictive intervals that contain the truth 2.8 % of the time. This pronounced undercoverage limits the reliability of downstream risk-sensitive decisions.

Adding the energy score raises coverage to 74.7 % and lowers CRPS by 25.9 % relative to the MSE-only LSTM (0.3377 to 0.2501), at a cost of 0.9 % in MAE; both energy-score variants still beat persistence on point accuracy by 8.5 %, so calibration is not purchased with accuracy. Fig. 2 compares empirical coverage at the nominal 90% level using the

numerical values in Table I. Notably, 74.7 % against a nominal 90 % remains under-calibrated by 15.3 percentage points; the improvement is substantial, although the intervals remain under-calibrated, and conformal or quantile recalibration is the natural next step.

B. *The adversarial term contributes nothing measurable*

Comparing T2 and GAN-LSTM isolates λ_a, since they differ only in that term: CRPS 0.2526 against 0.2501, coverage 70.7 % against 74.7 %, W_1 0.0238 against 0.0100. CRPS differs by 1.0 % and coverage by 4.0 points, but W_1 differs by a factor of 2.4, so these differences are not uniformly negligible; at a single seed, with the two variants differing by 4 points of coverage in one direction and 1 % of CRPS in the other, no consistent direction is identifiable. Multiple independent seeds are required to establish the stability of this comparison. Adding an adversarial term without a proper scoring rule (LSTM against T1) is a sharper comparison: point accuracy degrades by 7.5 % in MAE while spread improves, so the term is not inert but is not a substitute for a calibration objective either. Given how frequently GAN-based traffic generators are proposed, this remains an informative negative result.

C. *What the criterion predicts*

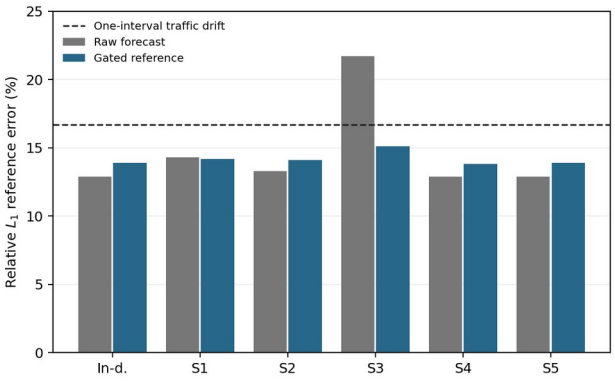


Fig. 3. Relative reference errors at latent rank 16, compared with one-interval traffic drift (dashed line), across all six scenarios.

TABLE II

MEAN DELAY (MS) BY REFERENCE MATRIX AT K = 16. OPTIMISER, PROPOSAL BUDGET, AND SEED ARE IDENTICAL ACROSS ROWS.

Reference matrix	In-d.	S1	S2	S3	S4	S5
d_{t-1} (reactive)	39.44	23.86	17.37	35.09	52.16	24.01
Raw forecast	36.95	23.77	16.43	41.53	51.11	23.96
Ensemble mean	35.74	25.04	14.98	39.00	51.92	23.03
Risk-weighted, ungated	34.58	23.59	16.42	40.37	46.46	21.46
+ drift gate	34.31	23.89	16.52	34.08	49.03	23.85
True d_t (oracle)	33.80	22.49	16.29	30.23	47.45	20.99

TABLE III

THE CRITERION ON BOTH SIDES. ONLY THE LATENT RANK CHANGES; THE TRAFFIC, AND THEREFORE THE DRIFT, IS IDENTICAL. DELAY IS AVERAGED OVER ALL SIX SCENARIOS.

	floor	ε_{fcst}	(1)	raw forecast	gated
$k=16$	11.4 %	14.3 %	holds	+0.9 % (1/6 lost)	−5.3 %
$k=8$	21.0 %	22.1 %	fails	+8.9 % (4/6 lost)	−3.2 %

Table II holds the optimiser fixed and varies only the matrix it is pointed at; Table III then varies the latent rank, which moves ε_{fcst} across the inequality while leaving $\delta_{drift} = 16.8\,\%$ untouched.

The observed result is narrower than the general criterion might suggest. The criterion does not predict the in-distribution sign: the raw forecast beats reactive routing in distribution in both configurations (36.95 against 39.44 ms at $k=16$; 37.97 against 39.44 ms at $k=8$). Reference error is a magnitude, and the routing outcome depends on the error's structure as well: a stale matrix errs in a temporally coherent direction that the routing configuration partially absorbs, so it can outperform a forecast even when the forecast has a smaller overall error.

The criterion is associated with a larger penalty under shift. With the inequality satisfied the raw forecast costs 0.9 % mean delay averaged over six scenarios and loses on one of them, its worst case being +18.4 % on S3, where the forecast error rises to 21.6 % and the inequality locally reverses (Fig. 3). With the inequality violated everywhere, the same controller costs 8.9 % and loses on four of six, with a worst case of +77.2 % on S2. The practical reading is therefore not that forecasting helps or hurts, but that these two quantities are cheap to measure, that their gap bounds the exposure a proactive controller is taking on, and that the exposure is realised under distribution shift rather than in steady state.

D. *The drift gate, and what it costs*

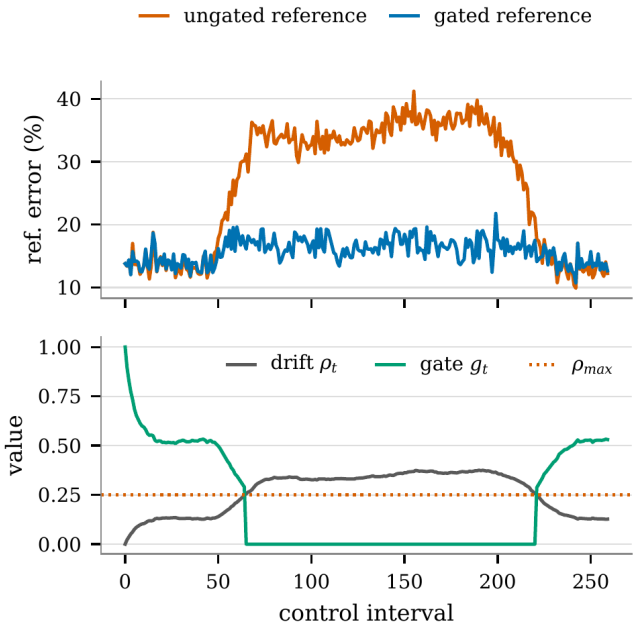


Fig. 4. The gate through the S3 volumetric surge. Top: reference error with and without gating. Bottom: the drift statistic rho_t crosses rho_max, the gate closes to zero for the duration of the surge, and reopens once the twin recovers.

The gate provides its largest benefit when the twin degrades. Under S3, ρ_t crosses ρ_{max}, the gate closes (Fig. 4), the reference error falls from 21.3 % to 15.7 %, and mean delay drops from 40.37 ms to 34.08 ms (15.6% relative to the ungated

risk-weighted controller) — turning the ungated controller's 15 % loss against reactive routing into a 3 % gain. Against reactive traffic engineering the gated controller gains 13.0 % in distribution (closing 90.9 % of the oracle gap), 4.9 % on S2, 2.9 % on S3, 6.0 % on S4 and 0.7 % on S5, and is 0.1 % worse on S1. It holds this advantage on both sides of the criterion, gaining 5.3 % at $k=16$ and 3.2 % at $k=8$ (Table III), which is the intended behavior of the fallback mechanism. The reported 90.9% in-distribution oracle-gap closure is (39.44 - 34.31)/(39.44 - 33.80), using reactive TE as the baseline and the oracle as the lower-delay reference (Table II).

Gating also introduces trade-offs relative to the ungated risk-weighted controller. Against the ungated risk-weighted controller one row above it in Table II, the gate wins decisively on S3 (-15.6 %) and marginally in distribution (-0.8 %), but loses on S4 ($+5.5$ %) and S5 ($+11.2$ %), for a mean across scenarios of -0.6 %. The gate therefore functions as a limited safeguard: it pays for itself where the twin fails badly and costs a few percent where the twin was fine, and its average benefit is dominated by the one scenario it was designed for. Its utility depends on the operating regime and the relative importance assigned to severe failures.

The weakest case is S1, the flash crowd, where exponential smoothing at $\beta=0.15$ registers an abrupt event too slowly: several intervals pass before the gate reacts, during which the controller still partly trusts a forecast that has not seen the event. A change-point detector rather than an EWMA is the natural fix, and this remains future work rather than a demonstrated capability.

E. Null result: DRL against classical local search

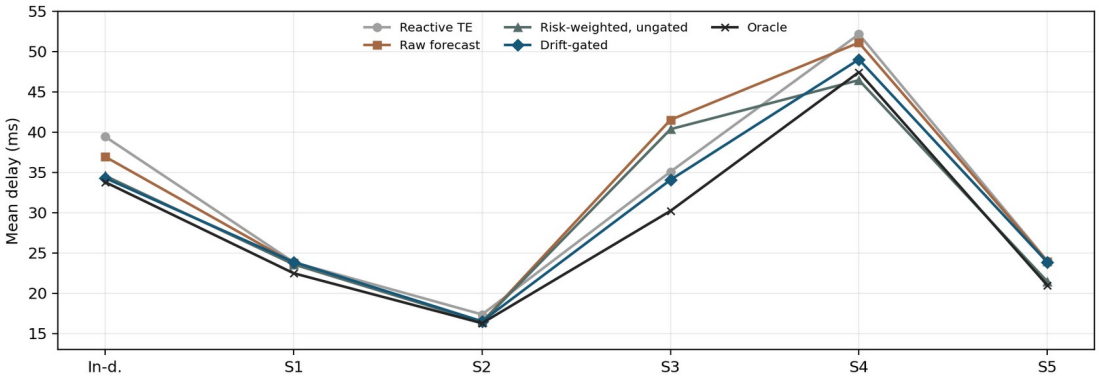


Fig. 5. Mean delay for five reference-matrix strategies across six scenarios. All plotted values are from Table II; the oracle is shown for comparison. The DRL results are discussed separately.

Fig. 5 plots the numerically reported delay values in Table II. In separate DRL runs, the PPO variants approached static-OSPF performance and were outperformed by classical local search. The three agents differ in whether they observe the twin at all, in which twin conditions them, and in whether the CVaR risk premium and adversarial curriculum are active, yet all three land within a few percent of the same operating point. The policy standard deviation contracted only from 0.200 to 0.197 over 400 episodes, against a contraction to 0.056 that the same optimiser achieves on a benchmark with a known optimum — the policies are converging, but to the identity action.

An action-space mismatch is a plausible explanation, rather than an established optimiser defect. Shortest-path routing is piecewise constant in the link weights: small perturbations change nothing until a path preference flips, at which point many link loads change discontinuously. A Gaussian policy exploring locally therefore sees a reward surface that is flat almost everywhere and discontinuous on a measure-zero set, providing gradient signal only through rare boundary crossings. Local search, which evaluates discrete candidates directly, is far better matched. This suggests the productive formulation is discrete selection — choosing among candidate configurations, or choosing which reference matrix to optimise against — while leaving continuous optimisation to a classical solver. The studied controller is such a hybrid: the learned component supplies the reference, the classical component supplies the optimisation.

VIII. Discussion and Limitations

Two consequences follow from (1). First, because ε_{fcst} is bounded below by representation loss, the useful lever is often a richer representation rather than a better sequence model; halving the latent rank was enough to move the raw forecast's penalty from 0.9 % to 8.9 % without touching the forecaster. Second, when the inequality fails the right response is not to abandon the forecast but to shrink towards telemetry by a data-driven amount, which is what the gate does — and which recovers a net gain on both sides of the inequality.

Several limitations constrain interpretation. Traffic is synthetic, generated by a model with selected parameters; its assumptions are described for inspection, and (1) is stated so it can be re-evaluated on any trace, but replication on operational data remains the necessary next step. All results are on a single topology, NSFNET; denser topologies offer more path diversity and may affect the relative performance of traffic-engineering methods. Each configuration was trained at a single seed, so the small differences in the adversarial ablation should not be read as significant, and the differences cannot support a general directional conclusion. Coverage remains materially below nominal. The six scenarios are also not equally demanding or equally comparable: within the 400-interval evaluation window S1's flash occupies 9 % of intervals

and S5's failure 20 %, while S2 and S4 apply throughout, so a scenario in which little changes cannot show a large difference either way. S2 in particular is easier than the in-distribution week in absolute terms — a fresh gravity draw spreads load more evenly at the same aggregate volume — so its gain should be read as evidence that the gate detects a mis-specified basis, not that shifted traffic is harder to route. Finally, the DRL result is a statement about the formulation we implemented, not about DRL for traffic engineering in general.

The drift-gate parameters are fixed in the evaluated runs, but a validation procedure and a sensitivity sweep for the smoothing coefficient and discrepancy threshold are not documented. Their present settings should therefore be viewed as configuration choices, not validated universal thresholds. The controller also uses only eight ensemble rollouts; its selected worst-case tail contains two members, making the empirical CVaR reference a coarse risk proxy rather than a stable tail estimate. The 74.7% empirical coverage further limits risk-calibration claims. Repeated seeds, an independent threshold study and a second topology are necessary before broader reliability claims can be made.

IX. Conclusion

Proactive routing is not automatically better than reactive routing, and how much worse it can get is measurable in advance. Comparing one-step forecast error against one-interval traffic drift is inexpensive but not routinely reported in these evaluations; the comparison does not by itself predict the in-distribution sign, but the gap between the two bounds the exposure a proactive controller takes on, and that exposure is realised under distribution shift — a 0.9 % average penalty with one losing scenario when the inequality holds, against 8.9 % and four when it fails. Training the twin with the energy score raises the empirical coverage of nominal 90 % intervals from 2.8 % to 74.7 % and improves CRPS by 25.9 % at negligible cost in point accuracy, but the resulting ensembles remain under-calibrated. The twin–reality drift gate measures twin fidelity online and shrinks the optimiser's reference towards telemetry accordingly, improving on reactive traffic engineering in five of six scenarios and on both sides of the criterion, at the cost of a few percent on the scenarios where the twin was fine; under the gate-off condition the reference reverts toward recent telemetry. Future work should replicate on operational traces, replace the EWMA drift detector with a change-point test to handle abrupt events, and reformulate the learned controller as discrete selection over candidate configurations.

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